NGUYEN DUC TRUNG THANH

Java Backend Developer

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PROFESSIONAL SUMMARY

Java Backend Developer with nearly 5 years of experience, specializing in enterprise banking systems and high-accuracy credit transaction flows on the Backbase digital banking platform. Directly developed the credit transaction orchestration engine (dflow), enterprise onboarding, content management (CMS) and background jobs at TPBank for online guarantees, end-to-end guarantees, IDNES e-guarantee, online disbursement, credit-profile signing and unsecured lending. Proficient in integrating BPM, remote digital signing (SignHub, CA certificates), Enterprise Content Management (ECM) and core banking (FCC) over an ESB. Strong in designing multi-step state machines, maker-checker approval flows, idempotency handling to prevent duplicate transactions, asynchronous processing with retry, and performance tuning.

TECHNICAL SKILLS

Languages	Java 8, SQL, JavaScript
Platforms & Frameworks	Spring Boot 2.2.x, Spring Data JPA/Hibernate, Spring AOP, Spring Scheduling, Backbase Building Blocks / Service SDK 10.1.0, Netflix Eureka, RESTful API, Microservices architecture, Struts 1/2
Core Banking & Integration	ESB / Core banking FCC (Flexcube), BPM Workflow Engine, remote digital signing (SignHub, CA certificates, RA), Enterprise Content Management (ECM), OTP flow, eKYC, IDNES/EGP e-guarantee partner, Webhook/Callback architecture, asynchronous processing with retry
Databases & Caching	Oracle DB, MS SQL Server, MySQL, Tomcat JDBC Pool
Document Processing & Security	Apache POI, PDFBox / openhtmltopdf, Thymeleaf, Velocity, BouncyCastle (digital signing/CA), internal JWT
Messaging, Async & Scheduling	Apache Kafka, Thread pool, multi-tier background schedulers, ActiveMQ
DevOps, Tools & Testing	Maven, Docker, Jenkins (CI/CD), Git, Jira, SonarQube, JaCoCo, JUnit 4, Mockito, PowerMock

WORK EXPERIENCE

TPBank — Backend Developer *March 2024 – Present*

1. Online Guarantee

Tech stack: *Java, Spring Boot, Backbase, Oracle DB, SignHub (digital signing/CA), ECM, BPM, ESB/FCC*

- Developed the online bid-guarantee issuance flow for corporate customers: create, edit and cancel transactions, search bid packages and beneficiaries, and manage drafts.
- Implemented a role-based multi-step signing state machine (SignRoleValidator): draft → OTP verification → chief accountant signing → legal representative signing → CA digital signing → processing, transitioning states according to product configuration (BLDT/DXCCKS).
- Integrated SignHub remote signing and generated digitally signed guarantee letters (PDF) from templates (Apache POI, PDFBox, Velocity), uploaded them to ECM and mapped ecmFileId; handled signing retry on failure via a counter mechanism.
- Sent notifications to customers at key status milestones of the guarantee transaction.

2. End-to-End Online Guarantee

Tech stack: *Java, Spring Boot, Backbase, Oracle DB, BPM, ECM, ESB/FCC*

- Developed a full-scope end-to-end guarantee flow: bid guarantee, warranty guarantee, advance-payment guarantee, contract-performance guarantee, payment guarantee, final-settlement guarantee, and financial commitment.
- Built the transaction initiation validation: verifying guarantee-letter details, beneficiary, joint-venture information and the debit account to enforce business constraints before entering the signing flow.
- Orchestrated transaction submission to BPM and accounting via ESB to core banking FCC; handled two-way callbacks and states BANK_RECEIVED, BPM_SUCCESS, COMPLETE, REJECT, CANCEL with asynchronous processing and full audit logging.
- Sent result-update notifications to customers according to each transaction state.

3. IDNES/EGP E-Guarantee Integration

Tech stack: *Java, Spring Boot, Backbase, Oracle DB*

- Built the public integration-api connecting the IDNES/EGP e-guarantee partner: issue (create), cancel, update, extend, relieve, claim and reconciliation of electronic guarantee certificates.
- Implemented e-guarantee transaction initiation validation, partner access-token issuance, and bank-branch directory lookup.
- Handled input security (checksum and token verification) and two-way DCB ↔ EGP transaction reconciliation to ensure data consistency and traceability.
- Sent notifications to customers upon partner responses (issue, update, cancel, claim).

4. End-to-End Online Disbursement

Tech stack: *Java, Spring Boot, Backbase, Hibernate JPA, ESB/FCC, Oracle DB*

- Developed the API to initiate end-to-end online disbursement transactions, linked to the credit contract, line serial, beneficiary information and attached documents.
- Built disbursement validation (loan term, disbursement date, principal/interest schedule) and integrated loan accounting and loan-account creation via ESB to FCC; recorded ESB details (fccRefNumber, error codes, message id) in the TRANSACTION table for reconciliation.
- Handled BPM callback, rejection and rollback flows, ensuring data consistency and full audit logging for each action.
- Sent notifications to customers when disbursement succeeded or failed.

5. Digital Credit-Profile Signing

Tech stack: *Java, Spring Boot, Backbase, Oracle DB, BPM Engine, SignHub (digital signing/CA), ECM*

- Developed the digital credit-contract/credit-profile signing flow, initiated from an external BPM via bpmTransCode with the SUBMIT_SIGN action, using bpmTransCode as the key to prevent duplicate transactions.
- Built a multi-layer pre-signing validator: verifying profile type, customer information (CIF), representative signer (card number), document count limit, and enforcing the signing deadline.
- Validated each signing document (ECM ID, allowed file type, visible signature position — page/offset); integrated CA digital signing via SignHub and synchronized signed documents with ECM, with asynchronous notification via a dedicated thread pool.
- Sent notifications to customers when the credit profile was signed successfully or rejected.

6. Unsecured Lending

Tech stack: *Java, Spring Boot, Backbase, Oracle DB, ESB/FCC, RESTful API*

- Developed the API to initiate unsecured (collateral-free) loan transactions for corporate customers, integrated into the shared dflow orchestration engine.
- Built the loan-statement lookup API and integrated loan-information queries via ESB to core banking FCC.
- Sent notifications to customers about unsecured loan status.

7. Enterprise Onboarding & Online Account Opening

Tech stack: *Java, Spring Boot, Backbase, Oracle DB, eKYC, VNeID, SignHub, ESB/FCC, AML/VMS, ECM, RPA, OCR/GenAI*

- Developed onboarding and eBank registration services for corporate customers (SME) and household businesses: registration journey, CIF/account creation, user and service-package creation, representative management, and profile export.
- Integrated eKYC electronic identity (document OCR, liveness, face matching, face search) and the VNeID enterprise information-sharing flow, together with OTP, captcha and video call (Stringee) in compliance with State Bank of Vietnam regulations.
- Integrated electronic contract signing via SignHub (FPT-CA, OTP signing verification); performed AML/VMS checks, business-license verification (VMG, RPA, GenAI-OCR) and account creation/blocking via ESB to FCC; protected endpoints with Zuul rate-limit and processed eKYC/VNeID flows asynchronously (Spring @Async).

8. Application Content Management (CMS)

Tech stack: *Java, Spring Boot, Backbase, Oracle DB, JPA Specification, ESB/FCC*

- Developed the CMS content-management module with full flows: create, search, view detail, and change content status (update/cancel/activate) by effective time, across multiple platforms and languages.
- Implemented the maker-checker approval flow: persisting create/status-change requests into the REQUEST table, checking permissions, and detecting conflicts with active content as well as pending requests to prevent duplication.
- Built a background job: scanning reminder records in INIT status by date, classifying by source system and message type (Trade Finance document-supplement reminders, locked-customer emails, due/overdue document emails), grouping by CIF/user and pushing notifications/emails via the notification service, then updating SUCCESS/FAILED status.
- Ran the job on a shared BaseScheduler with a valid execution time window, multi-instance support, and a cron that activates CMS content when it becomes effective (check-change-cms every 5 minutes).

FPT Software — Software Developer *March 2022 – February 2024*

SingStat Mobile Application

Tech stack: *Java, Struts 2, Spring Framework, Hibernate, jQuery, Ajax, Oracle DB*

- Contributed to developing and maintaining data processing, aggregation and visualization modules for the SingStat application of the Singapore Department of Statistics, serving citizens and government agencies looking up socio-economic figures.
- Built and optimized backend APIs querying large statistical datasets on Oracle DB, improving response time and stably handling high daily concurrent traffic.
- Analyzed business requirements, fixed bugs and added features per public-sector client feedback, ensuring the accuracy of published data.
- Wrote unit tests and technical documentation and took part in code reviews to maintain code quality and high test coverage for production releases.

Material Transfer Management System

Tech stack: *Java, Struts 2, Hibernate, MS SQL Server, jQuery*

- Developed a material transfer management system for a Japanese manufacturing enterprise: inventory tracking, inbound/outbound/inter-warehouse processes, and material transfer between departments.
- Built import/export request validation and approval flows and material data reconciliation between warehouses, ensuring data integrity.
- Developed an automated report generation engine (daily/monthly) and data-export features for operations and internal audit.
- Optimized queries and designed indexes on MS SQL Server for large material data tables; handled and fixed issues arising during operation.
- Coordinated with the Japanese client to clarify requirements (via documents and a bridge engineer), ensuring delivery matched specifications and quality standards.

4M Manufacturing Automation System

Tech stack: *Java, Spring Boot, MySQL, REST API*

- Contributed to developing manufacturing execution system (MES) modules for the Meiko factory, supporting monitoring and automation of the production process under the 4M model (Man, Machine, Material, Method).
- Built a lot-code generation algorithm and real-time material-lot traceability features, enabling origin tracing and product quality control on the production line.
- Designed and developed REST APIs connecting the application layer with production devices and databases, ensuring continuous and accurate data updates.
- Tested, debugged and optimized MySQL query performance for high-frequency production data recording.

EDUCATION

FPT University — Bachelor of Software Engineering
2020 – 2024

LANGUAGES

Vietnamese: Native | English: Professional working proficiency